

Embedded Systems Homework #3

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1 Days until new Years

```
1  #!/bin/bash
2
3  # Get the current year
4  current_year=$(date +%Y)
5
6  # Define New Year's Eve date (December 31)
7  new_years_eve="$current_year-12-31"
8
9  # Get the current date in the format YYYY-MM-DD
10 current_date=$(date +%Y-%m-%d)
11
12 a_day_in_seconds=86400
13
14 # Calculate the number of days until New Year's
   Eve
15 days_until_nye=$(( ( $(date -d "$new_years_eve"
   +%s) - $(date -d "$current_date" +%s) ) /
   $a_day_in_s >
16
17 # Display the result
18 echo "Days until New Year's Eve ($new_years_eve)
   : $days_until_nye days"
```

2 Compute Faculty

```
1  #!/bin/bash
2
3  while true; do
4      result=1
5
6      echo 'Enter a number (Ctrl+C to exit):'
7      read number
8
9      if [ $number -eq 0 ]; then
10         echo 1
11     elif [ $number -lt 0 ]; then
12         echo "Number can't be negative"
13     else
14         while [ $number -gt 1 ]; do
15             result=$((result * number))
16             number=$((number - 1))
17         done
18         echo -e "Factorial: $result\n"
19     fi
20 done
```

3 Passwd User List

```
1      #!/bin/bash
2
3      # Specify the path to your ASCII text file
4      file_path="/home/luffy/../../etc/passwd"
5      counter =1
6
7      # Check if the file exists
8      if [ -e "$file_path" ]; then
9          # Open the file for reading
10         while IFS= read -r line; do
11             # Extract the username at index 0 (the first
12              field) using awk
13             username=$(echo "$line" | awk -F':' '{print
14              $1}')
15             # Print the extracted username
16             echo "The $counter account is $username"
17             counter=$((counter + 1))
18         done < "$file_path"
19     else
20         echo "File not found: $file_path"
21     fi
```